

Instructions

There are 15 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

Question 1 (Suggested maximum time: 5 minutes)

Question 1 (Suggested maximum time: 5 minutes)

(a) Find the value of each of the following.

(i)	$35 - 16$	$=$	
------------	-----------	-----	--

(ii) $4 \times (6 + 8)$ =	
----------------------------------	--

(iii) $37.5 \div 2.5$	=
-----------------------	---

(iv) $\sqrt{11 \cdot 56} + 5^3$	$=$
--	-----

A large rectangular grid of graph paper, consisting of 20 columns and 10 rows of squares. The grid is empty and occupies the majority of the page below the header.

(b) Using only the numbers in the following list,

12, 35, 9, 19, 44

(i) write down two numbers that **add** up to 53,

and

(ii) write down a **prime** number,

Answer:

(iii) write down a **multiple** of 7,

Answer:

--

(iv) work out the **Lowest Common Multiple** of 12 and 44.

Answer:

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Question 2 (Suggested maximum time: 15 minutes)

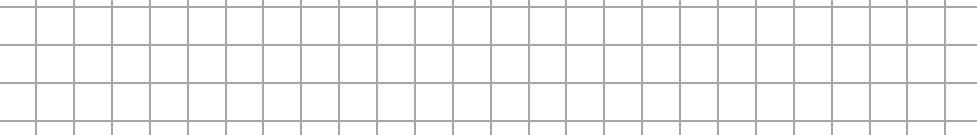
Darren has €850 to spend at a bicycle shop. The following items are on his shopping list:

Item	Cost (€)
Bicycle	605
Bicycle helmet	55
Cycling jersey	68
Bicycle wheel reflector	11



All the prices listed above include VAT at 21%.

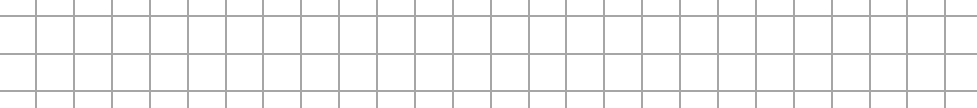
- (a)** Darren buys a bicycle, a bicycle helmet and two cycling jerseys. Work out the total cost of these items.



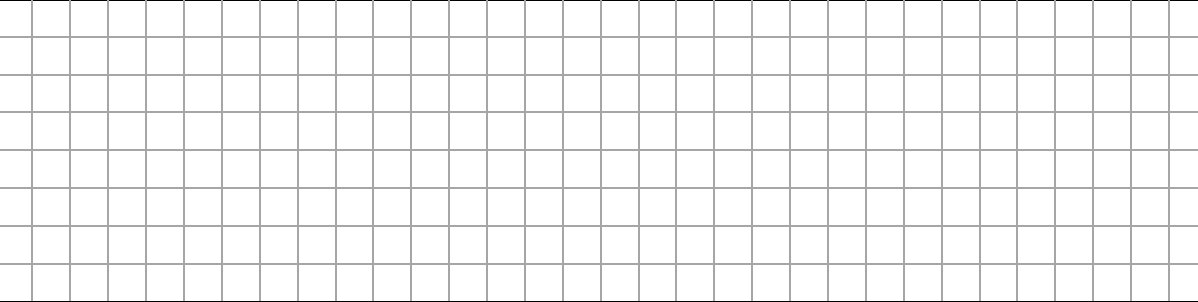
- (b)** Work out how much money he has left.

[illegible]

- (c)** Work out how many bicycle wheel reflectors Darren can buy with the money he has left.



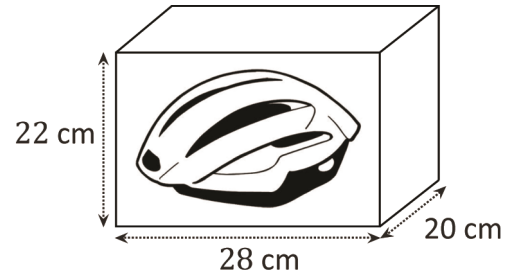
- (d)** Work out the cost of the bicycle **before** VAT at 21% was added.

A large grid of graph paper, consisting of 20 columns and 10 rows of squares, intended for drawing a picture.

- (e) Darren entered the bicycle shop at 11:35. He completed his purchases and left the shop at 12:15. How long did he spend in the bicycle shop?



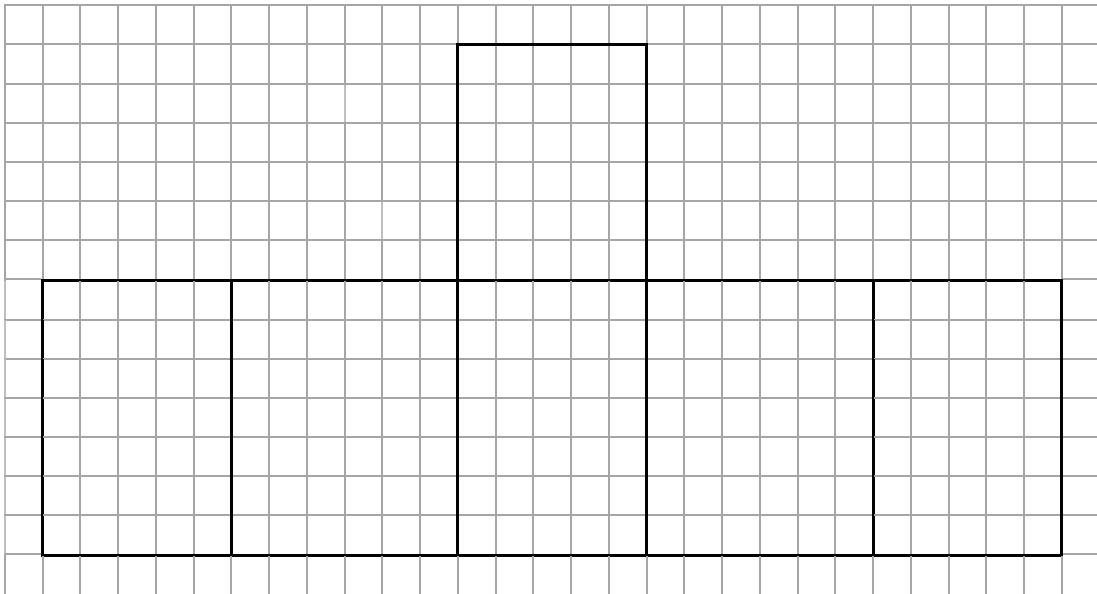
- (f) The bicycle helmet Darren purchased came in a rectangular box as shown in the diagram.



Work out the **volume** of the box, in cm^3 .

[illegible]

- (g)** Darren says that the net of a rectangular box is shown below. Is he correct?



Tick (✓) **one** box only. Give a reason for your answer.

Darren is correct Darren is **not** correct

9

7

Reason:



(Suggested maximum time: 5 minutes)

-

(i) Arrange the heights in ascending order.

--	--	--	--	--	--	--

(ii) The **median** height of the footballers is:

1.9

1

10

1

--	--

- The **probability** that the footballer's height is greater than 1.85 m is:

$$\frac{2}{7}$$
$$\frac{3}{5}$$
$$\frac{5}{7}$$
$$\frac{3}{7}$$

1

1

1

--	--

-
- A horizontal number line with arrows at both ends. It is marked with tick marks at intervals of 0.1, labeled from 0 to 1. The labels are: 0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.

- (i) Which number on the number line above has the same value as $\frac{1}{2}$?

[illegible]

- (ii) Which number on the number line above has the same value as $\frac{2}{5}$?

[illegible]

- (iii) Show the approximate position of $\frac{3}{4}$ on the number line above.

[illegible]

Question 4 (Suggested maximum time: 10 minutes)

Question 4 (Suggested maximum time: 10 minutes)

- (a)** Katie earns €2700 per month. She pays tax at 20%.

- (i)** Find 20% of €2700.

Katie has a tax credit of €137.50 per month.

- (ii) Work out Katie's **net tax** per month.

- (iii) Work out Katie's **net pay** per month.

[illegible]

- (b)** Katie received a pay rise. She is now earning €3100.

- (i) Work out how much Katie received as a **pay rise**.

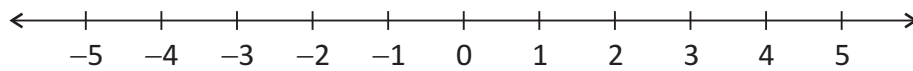
[illegible]

- (ii) Work out the **percentage increase** in Katie's pay.
Give your answer correct to the nearest integer.

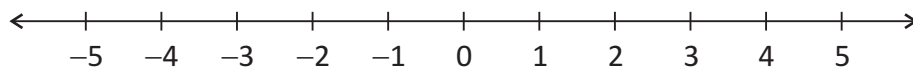
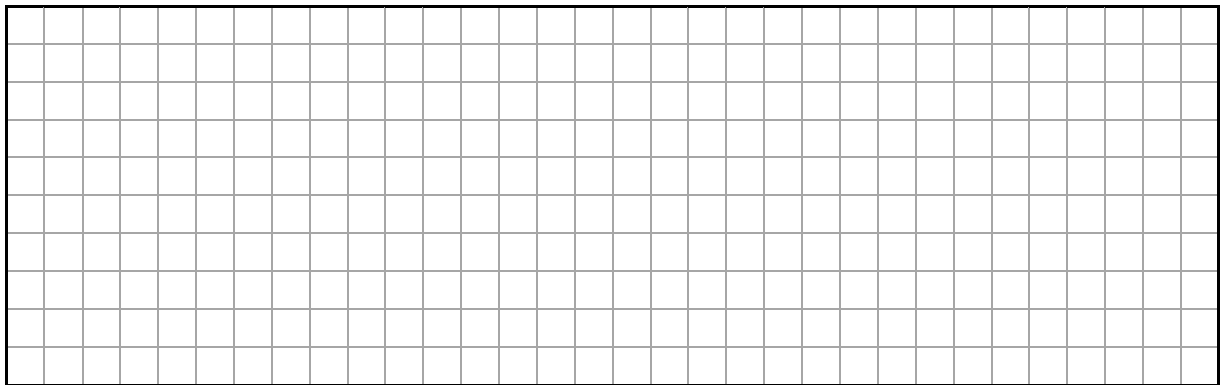
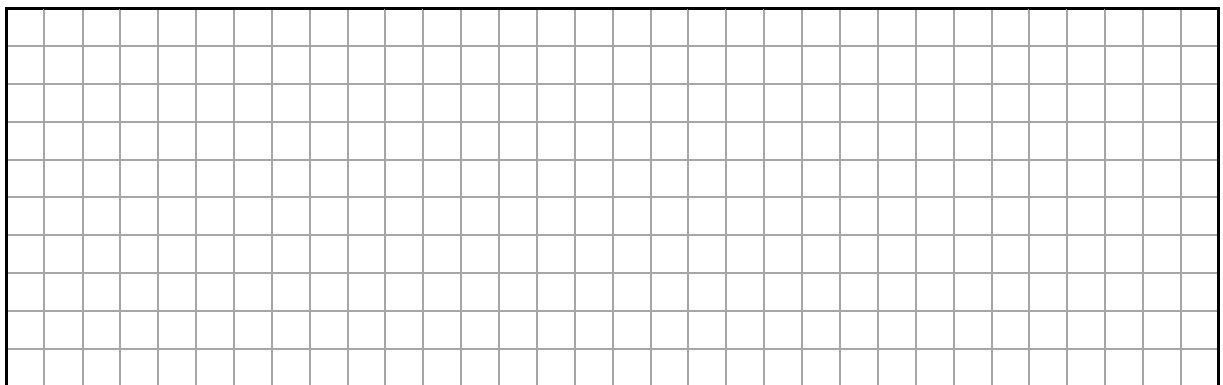
[illegible]

Question 5**(Suggested maximum time: 5 minutes)****(a)** Graph each of the following inequalities on the number line given.

(i) $x \leq 4, x \in \mathbb{Z}$.



(ii) $x < 1, x \in \mathbb{Z}$.

**(b)** Find the value of $3x - 4y$, when $x = 7$ and $y = 3$.**(c)** Multiply out and simplify $x(3x - 6) + 2(x + 4)$.

(Suggested maximum time: 10 minutes)

23, 29, 14, 6, 28, 9, 25, 30, 18, 14, 35.

- | | | | | | | |
|---|--|--|--|--|--|--|
| 0 | | | | | | |
| 1 | | | | | | |
| 2 | | | | | | |
| 3 | | | | | | |

2

3

$$=$$

- Number of students =

- Probability =

- Mode =

- [illegible]

-
- A large grid of graph paper, consisting of 20 columns and 10 rows of squares, intended for drawing a picture.

- How will this affect the **mean** (average)?
Tick (✓) **one** box only. Give a reason for your answer.

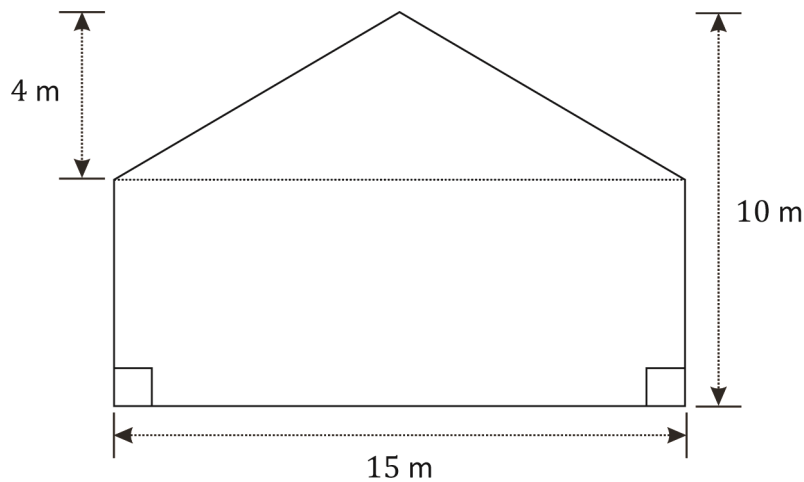
5

Reason: _____

Question 7

(Suggested maximum time: 5 minutes)

Sarah decides to paint the wall at the side of her house. The dimensions of the wall are shown below.



- (a)** Sarah says that the shape of the side wall is a combination of a **square** and a **triangle**. Is Sarah correct?

Tick (✓) **one** box only. Give a reason for your answer.

Sarah is correct

11

Sarah is **not** correct

7

Reason:

- (b)** Work out the **area** of the side wall of Sarah's house.



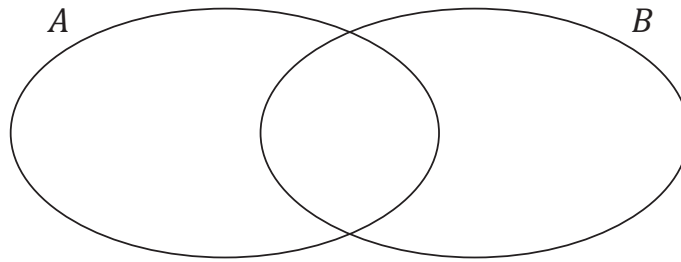
- (c)** One litre of paint covers 16 m^2 .
How many litres of paint will Sarah need to cover the side wall of her house?

[illegible]

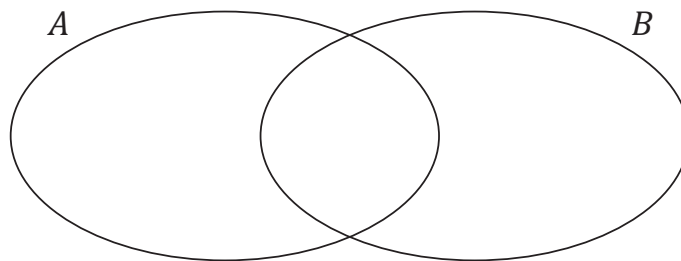
Question 8

(Suggested maximum time: 5 minutes)

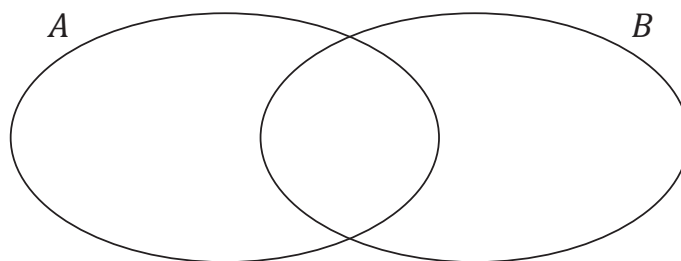
- (a)** On the Venn diagram below, **shade in** the region that represents $A \cup B$.



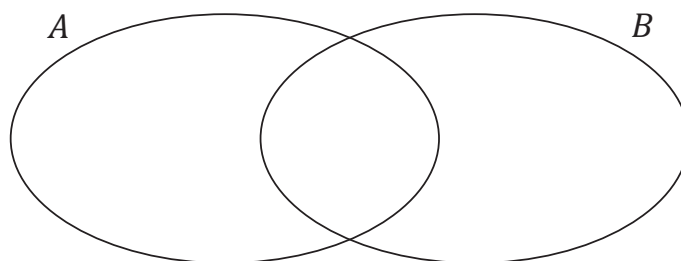
- (b)** On the Venn diagram below, **shade in** the region that represents $B \setminus A$.



- (c)** On the Venn diagram below, **shade in** the region that represents $(A \cap B)'$.



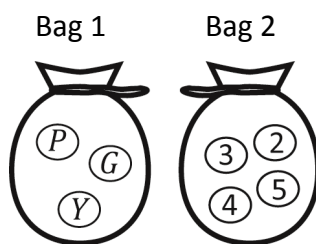
- (d)** B is a subset of A . Represent this, using **shading**, on the Venn diagram below.



Question 9

(Suggested maximum time: 5 minutes)

A game is played using two bags with counters as shown below.



Bag 1 contains a pink counter (**P**), a green counter (**G**) and a yellow counter (**Y**).

Bag 2 contains counters labelled 2, 3, 4 and 5.

A counter is drawn at random from Bag 1 and a counter is drawn at random from Bag 2.

Two possible outcomes are (P, 2) and (Y, 4).

(a) Complete the table to show all the possible outcomes.

		Bag 1		
		P	G	Y
Bag 2	2	(P, 2)		
	3			
	4			(Y, 4)
	5			

(b) How many possible outcomes are there?

Answer:

(c) Find the probability that the outcome contains a yellow counter (**Y**).

Probability =

(d) Find the probability that the outcome contains the number 3.

Probability =

(e) Find the probability that the outcome does **not** contain a yellow counter (**Y**) or the number 3.

Probability =

Question 10 (Suggested maximum time: 5 minutes)

Question 10 (Suggested maximum time: 5 minutes)

(a) Factorise fully each of the following.

(i) $2ab - 8b$

$$2ab - 8b = 2b(\quad)$$

(ii) $x^2 + x - 12$

$$x^2 + x - 12 = (x \quad)(x \quad)$$

(iii) $pq - 2p + 3q - 6$



(b) (i) Factorise fully $t^2 - 25$.

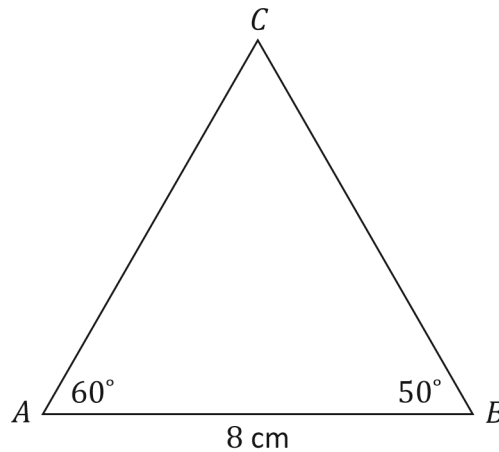
[illegible]

(ii) Using your answer from part (i), or otherwise, solve the equation $t^2 - 25 = 0$.

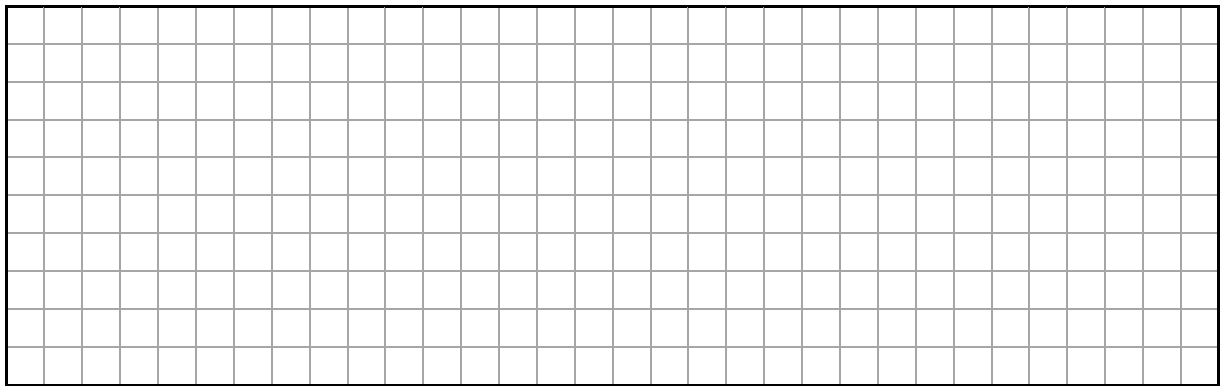
[illegible]

Question 11**(Suggested maximum time: 10 minutes)**

The sketch shows the triangle ABC , with $|AB| = 8\text{ cm}$, $|\angle CAB| = 60^\circ$ and $|\angle ABC| = 50^\circ$.



- (a)** Work out the size of the **third angle** of the triangle ABC .



- (b)** Put a tick (✓) in the correct box to show what type of triangle is shown in part **(a)** above. Tick **one** box only. Give a reason for your answer.

Isosceles

☐

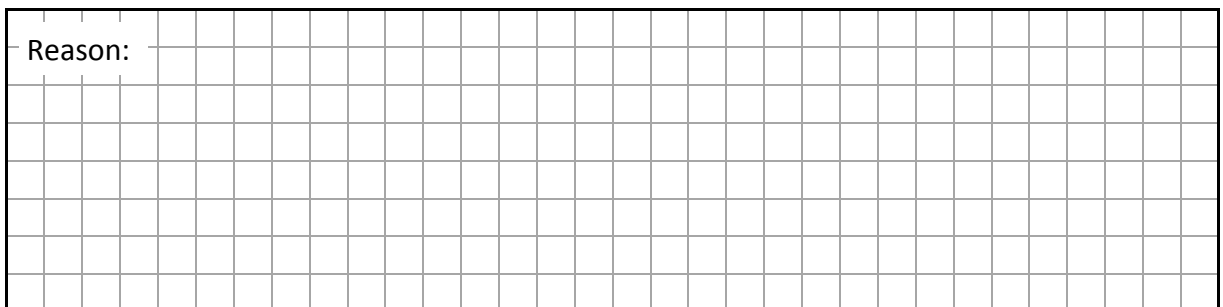
Equilateral

☐

Scalene

☐

Reason:



- (c) **Construct** an accurate drawing of the triangle ABC in the space below.
Show all of your construction lines clearly.



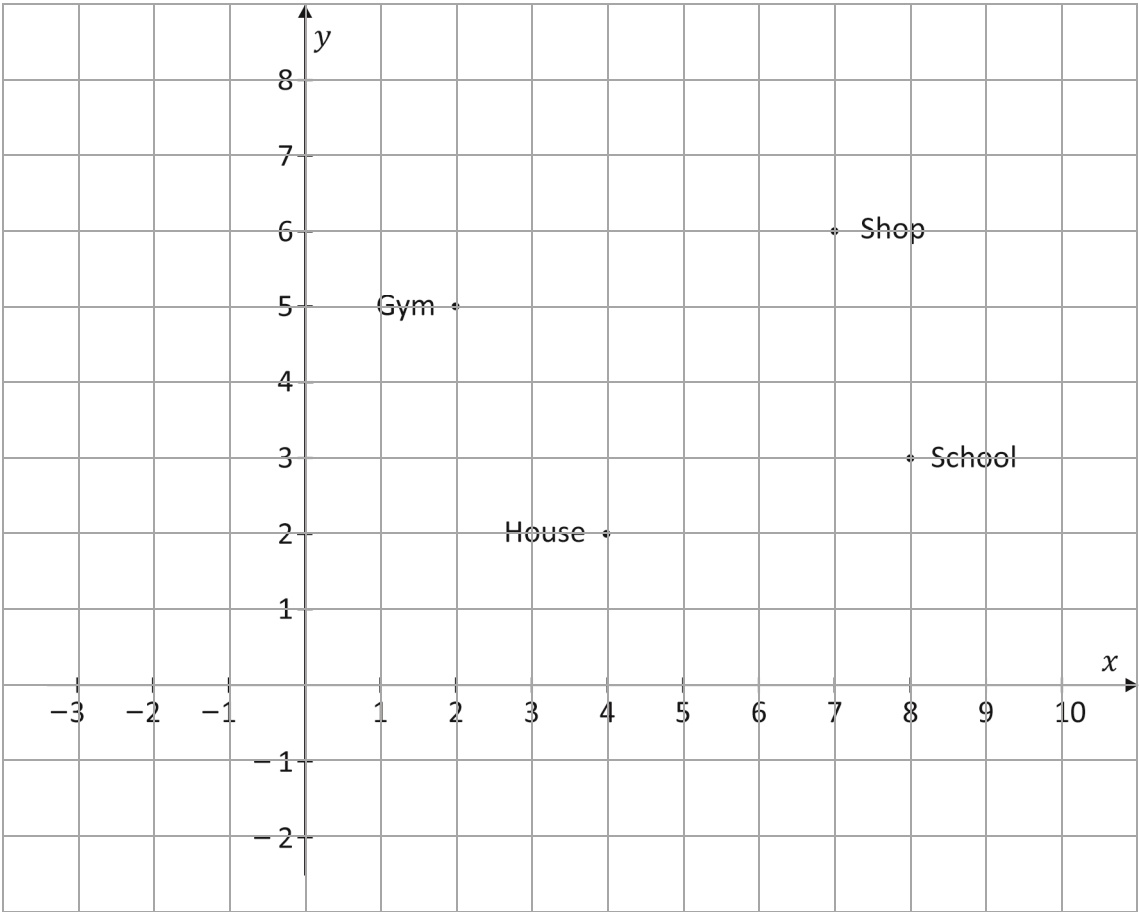
- (d) From your construction, measure $|AC|$.
Give your answer correct to one decimal place.

$|AC| =$ cm

Question 12

(Suggested maximum time: 10 minutes)

The co-ordinated diagram below shows the position of four buildings near Sian’s house.
Each small square on the diagram has a side of length 1 cm.

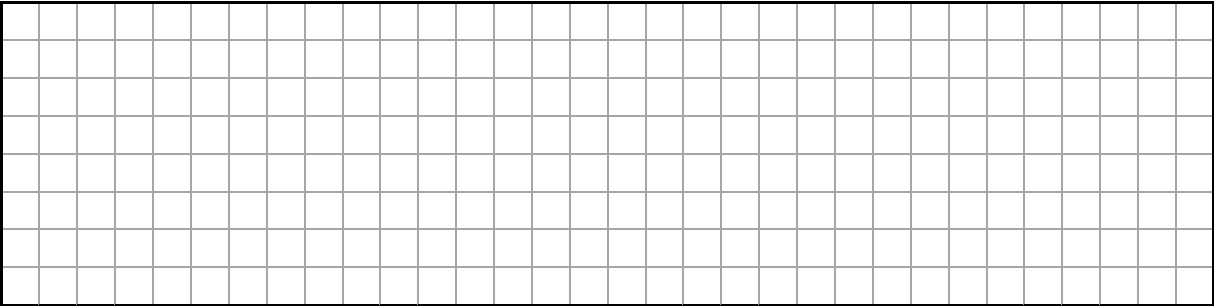


(a) Complete the table below to show the co-ordinates of the four buildings.

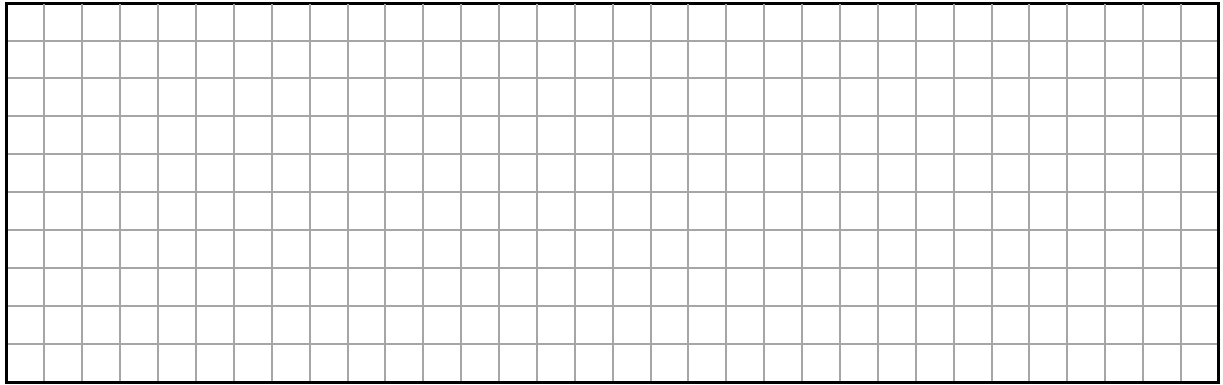
Point	Home	School	Shop	Gym
Co-ordinates	(4, 2)	(,)	(,)	(,)

(b) A church is located midway between the school and the gym.
Find the co-ordinates of the location of the church.

Co-ordinates of church = (,)



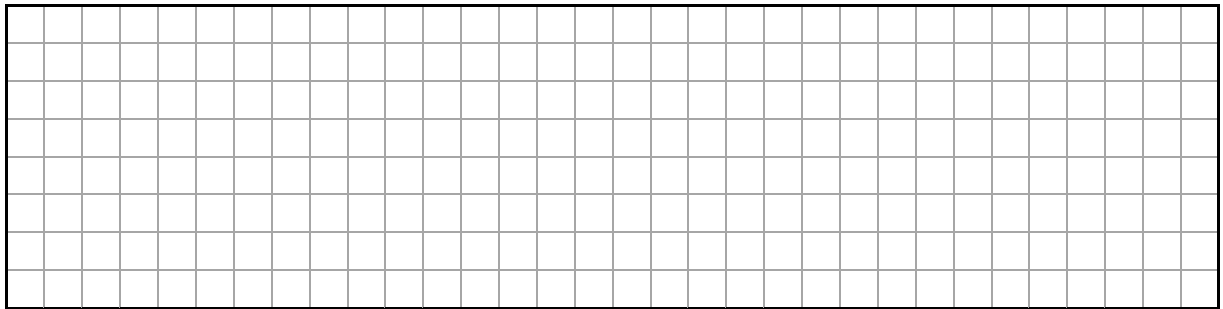
- (c) Work out the distance between Sian's home and the shop.



- (d) The co-ordinated diagram is to a scale of **1 cm = 150 m**.
Work out the actual distance between Sian's home and the shop.

Distance between Sian's home and the shop =

m



Question 13 (Suggested maximum time: 10 minutes)

Question 13 (Suggested maximum time: 10 minutes)

- (a)** $\{(3, 9), (4, 12), (5, 15), (6, 18)\}$ are couples of the function h .

- (i) Write down the elements of the **domain** and the **range** of the function h .

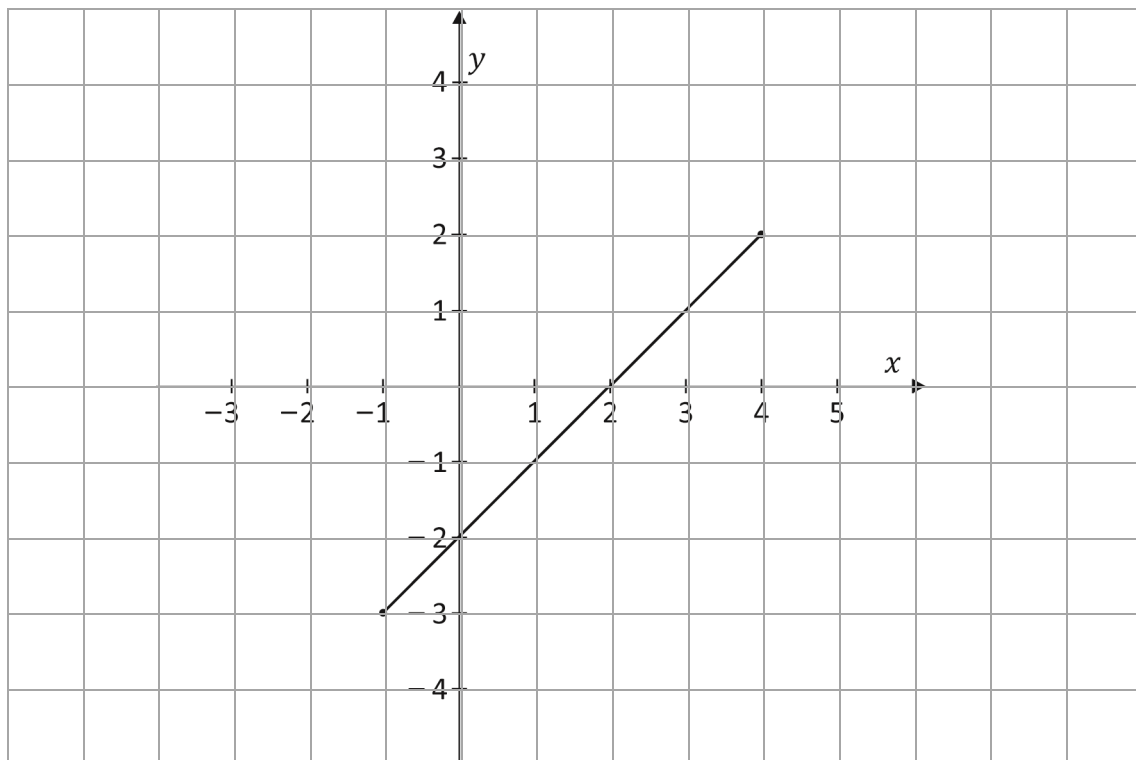
Domain = $\{ \quad \quad \quad \}$

Range = { }

- (ii) Explain the relationship between the **domain** and the **range** of the function h .

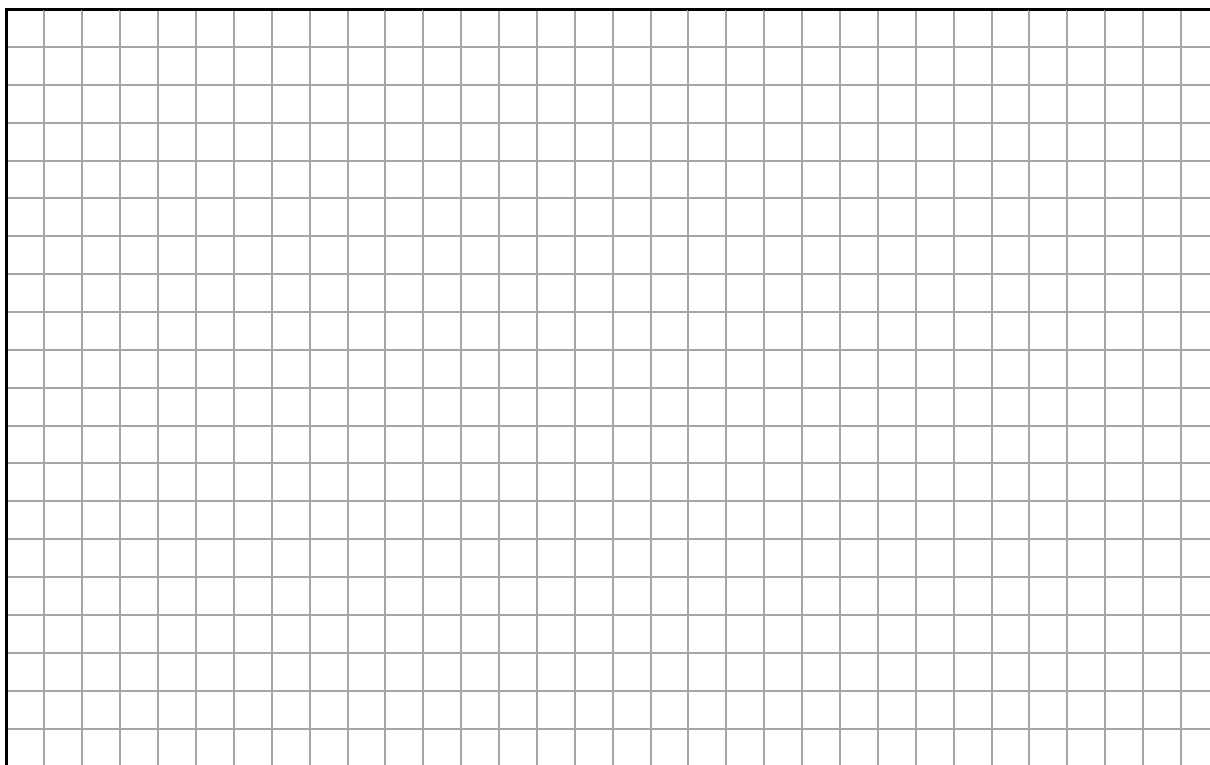
[illegible]

- (b)** The co-ordinated diagram below shows the graph of the function $f(x) = x - 2$.



- (i) On the same axis on the previous page, draw the graph of the function

$$g(x) = x^2 - 3x - 1 \text{ for } -1 \leq x \leq 4, x \in \mathbb{R}.$$



- (ii) Use the graph to estimate the points of intersection of $f(x)$ and $g(x)$.
Give your answers correct to one decimal place.

(,)

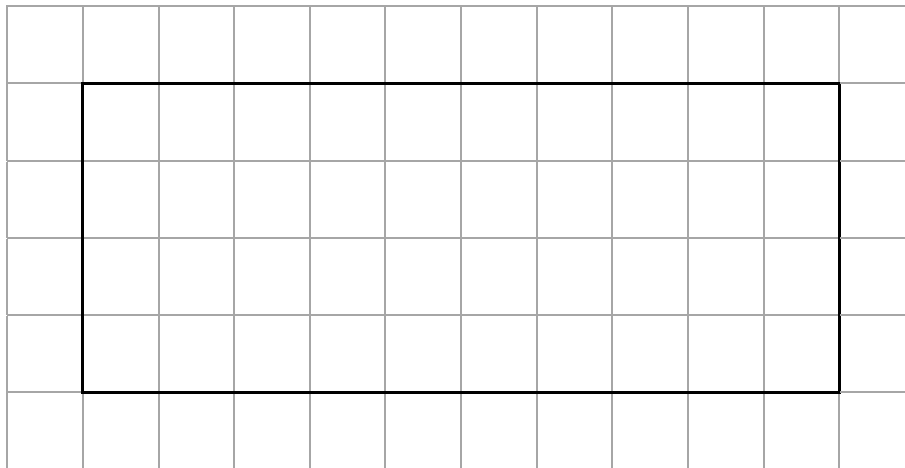
and

(,)

Question 14

(Suggested maximum time: 5 minutes)

Sean constructs a model of his garden as shown in the diagram below.
The model is rectangular in shape and measures 10 cm by 4 cm.

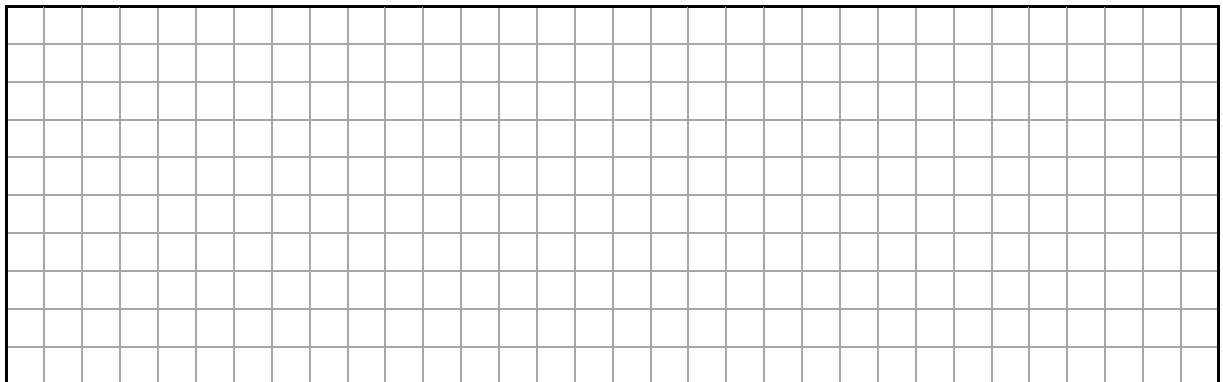


- (a) Draw in the two axes of symmetry on the diagram above.
- (b) Draw a **diagonal** on the diagram of the garden above.
- (c) Measure the length of the diagonal you have drawn in part (b).
Give your answer correct to one decimal place.

Length of the diagonal =

cm

- (d) Use the theorem of **Pythagoras** to confirm your answer in part (c).



Question 15

(Suggested maximum time: 5 minutes)

Liam and Olivia saved money for their holidays.

Liam saved €6 more than Olivia.

Olivia's savings in euro (€) are represented by x .

- (a) Select one suitable expression from the list below which represents the amount of money that Liam saved in euro (€).

$6x$	$x - 6$	$x + 6$	$6x + 6$
------	---------	---------	----------

Expression:

--

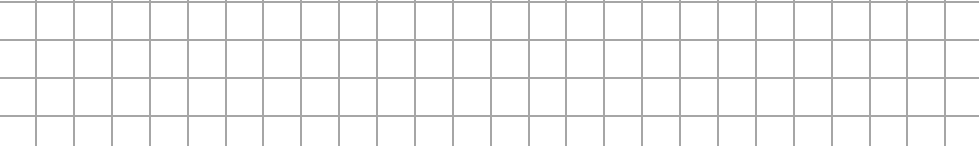
- (b)** Olivia said that the total amount that she and Liam have saved between them can be represented by the expression $2x + 6$.

Show that she is correct.

- (c) The total amount of money that Liam and Olivia saved was €82. Write down an equation in x to represent this statement.

[illegible]

- (d)** Solve your equation to find the amount of money that Olivia saved.



Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

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Pre-Junior Cycle Final Examination, 2022

Mathematics – Ordinary Level

Time: 2 hours

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